

PhD Math Camp

Calculus II: Integrals

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Outline

From Probability to Areas

Riemann Integration

Antiderivatives and the Fundamental Theorem

Distributions and Applications

Numerical Integration

From Probability to Areas

Introduction

- In this lecture we will learn what **integrals** are and how are they useful.
- But we will motivate integrals from a specific angle: **probability**.
 - We will learn that integrals help compute probabilities for continuous variables.
 - Also, for continuous variables, integrals define important properties (aka, *parameters*). For example, the true mean of a population is defined with an integral.
- Integrals show up in many applications, but fortunately computers will do the calculations. **We just need to know what an integral is, and what does it mean when it shows up in a specific application.**

Sample Space, Events, Random Variables, and Probability

- Sample space, Ω , includes all possible outcomes of a random process.
- An event is a subset of the sample space, e.g., all elements $x_i \in \Omega$ where a property $Q(x_i)$ is true.
- A random variable (RV), X , is numerical and represents a collection of outcomes of a random process (an event). X can be discrete (integer subset) or continuous (real subset).
- The probability of an event is the count of favorable outcomes (i.e., the event happens) divided by all outcomes.
- Generally, for any discrete RV, X , then

$$P(X = x_i) = \frac{|\{\omega_j \in \Omega : \omega_j \leftrightarrow X = x_i\}|}{|\Omega|} = \frac{\text{\#Favorable Outcomes}}{\text{\#All Outcomes}}$$

Discrete vs. Continuous Random Variables

- **Discrete Random Variable (DRV):**

- Let X be a DRV, then X takes on values from a subset of integers, $\mathbb{Z}$.
- Probability function is defined as, $f : X \rightarrow [0, 1]$, such that for each $x_i \in X$, the probability function $f(x_i)$ gives the probability that X equals x_i .

- **Continuous Random Variable (CRV):**

- Let X be a CRV, then X takes on infinitely many values from a real subset.
- Probability function is defined as, $f : C \subseteq X \rightarrow [0, 1]$, such that for any subset $C = [a, b] = \{x_i \in X : x_i \in [a, b]\}$, the probability function $f(C)$ yields the probability that X lies in C .

From the Probability Distribution Function to the Cumulative Distribution Function (CDF)

The CDF:

- The Cumulative Distribution Function (CDF) calculates the probability of obs. x_i or a lower value. It sums all probabilities up to x_i . For a Discrete RV:

$$\Pr(X \leq x_i) = CDF(x_i) = F(x_i) = \sum_{k=-\infty}^i f(x_k)$$

- Thus, starting from a Probability Distribution Function, we can always construct the Cumulative Distribution Function by adding up the probabilities from the minimum possible value of X up to the probability of x_i

From the Cumulative Distribution Function (CDF) to the Probability Distribution Function

Deriving Probability from CDF:

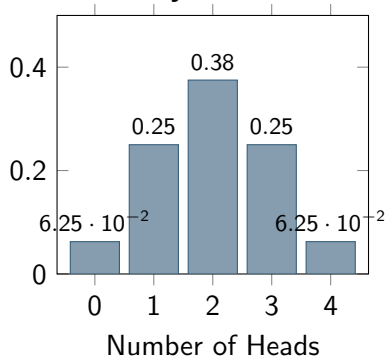
- With the CDF of a random variable, you can find its Probability Distribution Function.
- For a Discrete RV, simply subtract the CDF values at two successive points to find the probability of the latter point:

$$F(x_i) - F(x_{i-1}) = \sum_{k=-\infty}^i f(x_k) - \sum_{k=-\infty}^{i-1} f(x_k) = f(x_i)$$

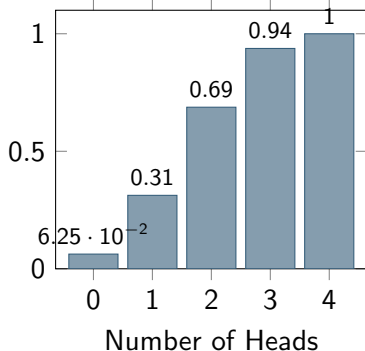
- This allows you to calculate probability at a point from accumulated probability.

Probability Distribution and Cumulative Distribution of the Total Number of Heads When Throwing Four Coins

Probability Distribution



Cumulative Distribution



Key Takeaway: Interplay of the Probability Distribution and the CDF

- When you look at a discrete Probability Distribution Function and CDF, the Probability Distribution at $x = a + 1$ equals exactly the rate of change in accumulated probability when one moves from a to $a + 1$ in the CDF.
- Metaphorically, the Probability Distribution seems to represent the “speed” at which we accumulate probability.
- Similarly, the CDF value at a given point a , is the sum of all probability “speeds” from the minimum possible value of X to a .
- This is a nice connection: **summing the areas under the Probability Distribution graph yields the CDF, and the “slope” of the CDF graph recovers the Probability Distribution.**

Continuous Random Variables and the Probability Density Function (PDF)

- *But*, continuous random variables have an important difference
 - **the probabilities are no longer points, but areas under a curve.**
- An infinitely thin slice in continuous space contributes no area, so to compute probabilities we use closed intervals, $[a, b]$, and a function, $f(x_i)$ that computes the *relative likelihood* of $x_i \in [a, b] \subseteq X$.
- We call f , the Probability **Density** Function (PDF). If f is the PDF of a Continuous RV X , then f needs to satisfy two conditions:
 1. The PDF is always non-negative, i.e. $f(x) \geq 0$ for all x in the domain of X .
 2. The area under the entire PDF curve is 1.

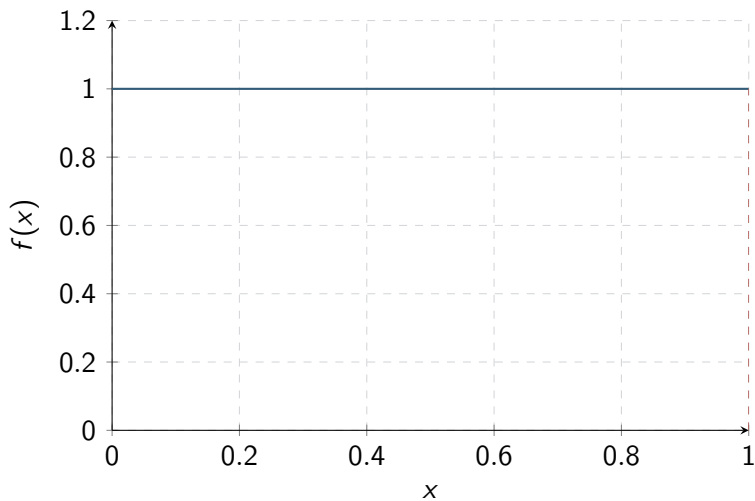
Uniform Distribution for Continuous Random Variables

- Consider the uniform distribution on the interval $[0, 1]$.
- The PDF is:

$$f(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- The total area under the PDF curve is 1.

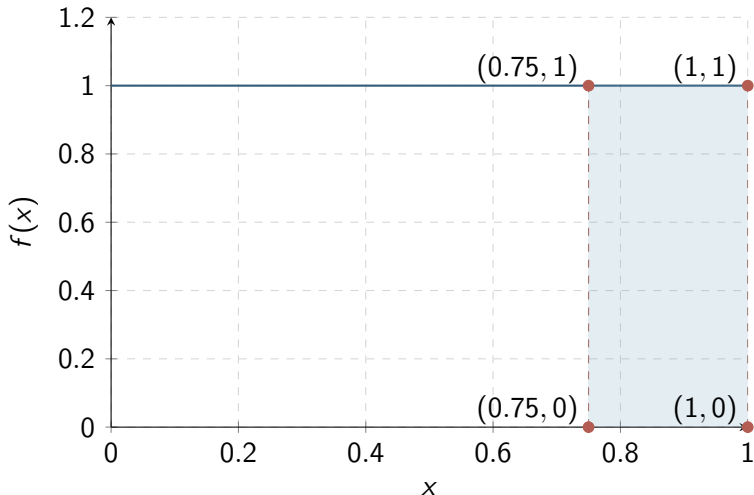
Graph of the Uniform Distribution PDF



Probability Calculation for Uniform Distribution

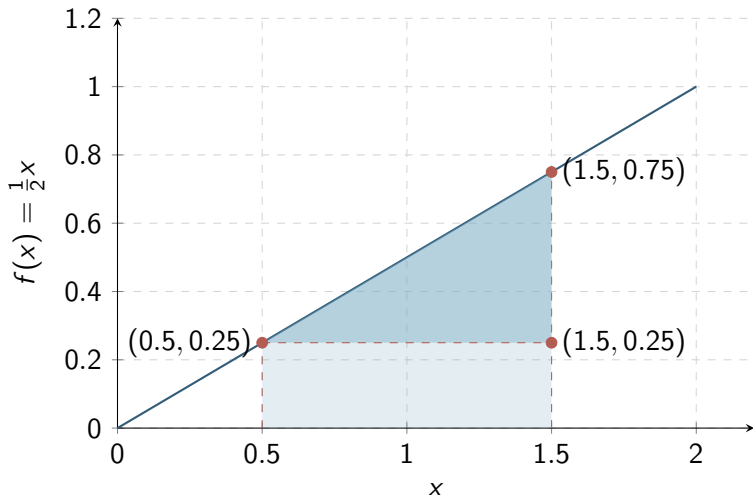
- The probability that X is greater than 0.75 in a uniform distribution from 0 to 1 is calculated as follows:

$$P(0.75 < X) = 1 \times (1 - 0.75) = 0.25$$



Another Example: a Triangular PDF

- Suppose $f(x_i) = (1/2)x_i \leftrightarrow x_i \in [0, 2]$ and $f(x_i) = 0 \leftrightarrow x_i \notin [0, 2]$.
- To calculate the probability that X lies between 0.5 and 1.5 we can use one rectangle and one triangle to represent the area.



How do we compute areas under a curve?

- So, given the previous definition, for any Continuous RV X with PDF $f(x_i)$, to calculate the probability that X lies in some interval $[a, b]$ we need to compute the area under the PDF curve between a and b .
- How do we do that for *any* curve $f(x_i)$?

Riemann Integration

Approximating Area: Riemann Sums

- **Riemann sums** are a method to approximate the area under a curve by breaking it down into simpler geometric shapes, such as rectangles.
- To compute a Riemann sum, we:
 1. Divide the interval $[a, b]$ into n subintervals of equal width, $\Delta x = \frac{b-a}{n}$.
 2. On each subinterval, construct a rectangle. The height of each rectangle is determined by the function value at a specific point within the subinterval.
 3. Sum the areas of all rectangles to get an approximation of the total area under the curve.

Approximating Area: Riemann Sums

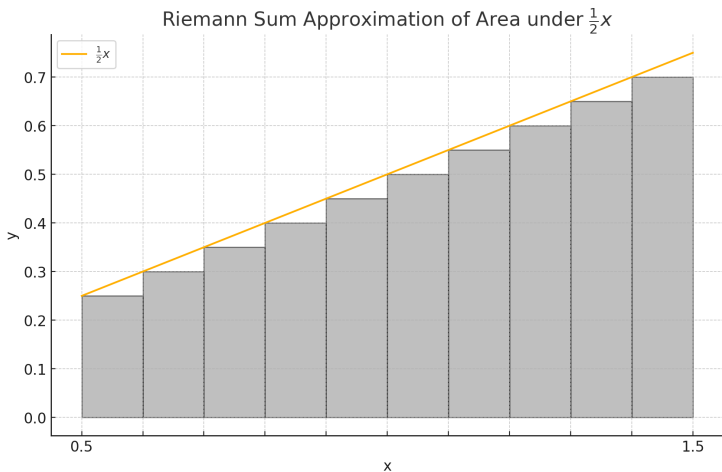
- The Riemann sum can be written as:

$$S_n = \sum_{i=1}^n f(x_i^*) \cdot \frac{b-a}{n} = \sum_{i=1}^n f(x_i^*) \Delta x$$

where x_i^* is a chosen point in the i -th subinterval.

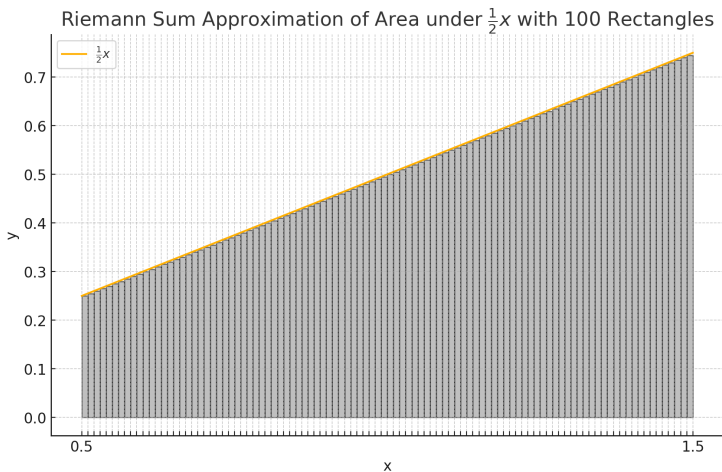
- As n increases (i.e., more rectangles), the approximation becomes more accurate, and in the limit as $n \rightarrow \infty$, the Riemann sum converges to the exact area, which is the integral of the function.

Triangular PDF: Riemann Approximation with 10 Rectangles



- Riemann Sum = 0.475. Exact Value = 0.50.

Triangular PDF: Riemann Approximation with 100 Rectangles



- Riemann Sum = 0.4975. Exact Value = 0.50.

The Riemann Integral: Taking the Limit

- While Riemann sums provide an approximation, the **Riemann integral** gives the exact area under the curve.
- The Riemann integral is defined as the limit of the Riemann sum as the number of subintervals n approaches infinity:

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x = \lim_{\|\Delta x\| \rightarrow 0} \sum_{i=1}^n f(x_i^*) \Delta x_i$$

- Here:
 - $\Delta x = \frac{b-a}{n}$ is the width of each subinterval.
 - x_i^* is a chosen point in the i -th subinterval $[x_{i-1}, x_i]$.
 - The sum $\sum_{i=1}^n f(x_i^*) \Delta x$ represents the total area of the rectangles approximating the area under $f(x)$.
- The integral $\int_a^b f(x) dx$ represents the exact area

Key Takeaway: Riemann Integrals and Their Challenges

- **Riemann integrals** provide a clear and intuitive way to understand the area under a curve by summing up the areas of rectangles and taking the limit as the number of rectangles approaches infinity.
- *Intuitive interpretation*: The integral represents the exact area under the curve $f(x)$ between two points a and b .
- **However**, calculating the limit of these sums to find the exact area can be tedious and complex, especially for complicated functions.
- This complexity raises the question: Is there a more efficient way to compute these areas without directly taking limits?

Antiderivatives and the Fundamental Theorem

Introduction to Antiderivatives

Def. (Antiderivative) An **antiderivative** of a function $f(x)$ is another function $F(x)$ whose derivative is $f(x)$.

- In other words, $F(x)$ satisfies:

$$F'(x) = f(x)$$

- *Example:* If $f(x) = 2x$, then one possible antiderivative is $F(x) = x^2$ because:

$$\frac{d}{dx} (x^2) = 2x$$

- **Key Idea.** The process of finding an antiderivative is often called *integration*, and it is the reverse of differentiation.

Antiderivatives and the Constant C

- When we find the antiderivative (indefinite integral) of a function $f(x)$, we obtain a **family of functions** rather than a single function.
- The general form of an antiderivative is given by:

$$F(x) + C$$

where C is an arbitrary constant.

Key Insight

The constant C accounts for the fact that differentiation of a constant yields zero, meaning any function of the form $F(x) + C$ has the same derivative $f(x)$.

- *Example:* For $f(x) = 2x$, the antiderivative is:

$$\int 2x \, dx = x^2 + C$$

where C could be any real number.

Common Rules for Finding Indefinite Integrals

Rule	Indefinite Integral
Power Rule	$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$
Constant Multiple Rule	$\int k \cdot f(x) dx = k \cdot \int f(x) dx$
Sum/Difference Rule	$\int (f(x) \pm g(x)) dx = \int f(x) dx \pm \int g(x) dx$
Exponential Rule	$\int e^{kx} dx = \frac{1}{k} e^{kx} + C$
Trigonometric Integrals	$\int \sin(x) dx = -\cos(x) + C$
	$\int \cos(x) dx = \sin(x) + C$
	$\int \sec^2(x) dx = \tan(x) + C$
Logarithmic Rule	$\int \frac{1}{x} dx = \ln x + C \quad (x \neq 0)$

The Fundamental Theorem of Calculus: Part 1

- The Fundamental Theorem of Calculus (FTC) connects differentiation with integration, offering a practical way to compute integrals.

Thm. (FTC Part 1) If $F(x)$ is an antiderivative of $f(x)$ on an interval $[a, b]$, then:

$$\frac{d}{dx} (F(x)) = \frac{d}{dx} \left(\int_a^x f(t) dt \right) = f(x)$$

- This means that the derivative of the integral function of $f(x)$ (from a to x) is equal to the original function $f(x)$.
- **Key Insight.** Differentiation and integration are inverse processes.

The Fundamental Theorem of Calculus: Part 2

Thm. (FTC Part 2) If $F(x)$ is any antiderivative of $f(x)$, then the definite integral of $f(x)$ from a to b is:

$$\int_a^b f(x) dx = F(b) - F(a)$$

- **Key Insight.** Instead of using Riemann sums to find the area under $f(x)$, we can find an antiderivative $F(x)$ and evaluate it at the endpoints a and b .
- *Example:* If $f(x) = 2x$, then:

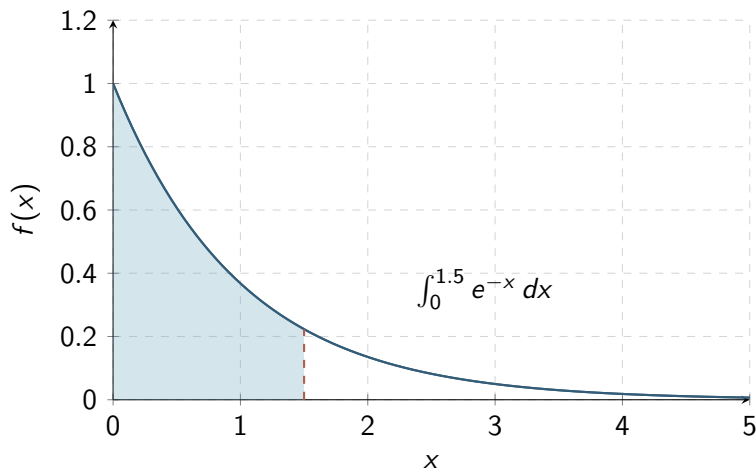
$$\int_0^2 2x dx = [x^2]_0^2 = F(2) - F(0) = 2^2 - 0^2 = 4$$

Distributions and Applications

Example: Exponential Distribution

- Consider the exponential distribution with rate parameter $\lambda = 1$:

$$f(x) = e^{-x} \quad \text{for } x \geq 0$$



Example: Computing Probabilities Using the Exponential Distribution

- Consider a random variable X that follows an **exponential distribution** with rate parameter $\lambda > 0$. The PDF is:

$$f(x) = \lambda e^{-\lambda x} \quad \text{for } x \geq 0$$

- Question:** What is the probability that X is less than or equal to a value a ?

$$P(X \leq a) = CDF(X) = \int_0^a \lambda e^{-\lambda x} dx$$

- Solution:**

$$\int_0^a \lambda e^{-\lambda x} dx = \left[-e^{-\lambda x} \right]_0^a = 1 - e^{-\lambda a}$$

- Thus, $P(X \leq a) = 1 - e^{-\lambda a}$.

The Normal Distribution

- **Normal Distribution:**

- Continuous probability distribution defined by mean μ and standard deviation σ .
- PDF:

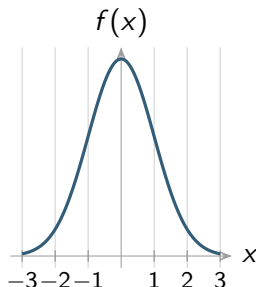
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- **Computing Probabilities:**

- Probability of an interval $[a, b]$:

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

- CDF: $F(x) = \int_{-\infty}^x f(t) dt$



Standard Normal Distribution

- **Standard Normal Distribution:**

- A special case of the normal distribution with mean $\mu = 0$ and standard deviation $\sigma = 1$.
- The probability density function (PDF) is given by:

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

- **Computing Probabilities using Integrals:**

- For $a \leq Z \leq b$, the probability is:

$$P(a \leq Z \leq b) = \int_a^b \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

- The CDF denoted by $\Phi(x)$, is expressed as:

$$\Phi(x) = P(Z \leq x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

Computing Probabilities for the Standard Normal Distribution

The area under the PDF from $-\infty$ up to x defines the $CDF(x) = P(Z \leq x)$.

- [Interactive Demo](#)

Derivatives, Integrals, and Probability

- **Calculating Changes Over Intervals:**

- To find the change in any $F(x)$ from $x = a$ to $x = b$, just compute $F(b) - F(a)$.
- If $F(x)$ is the CDF, then
$$F(b) - F(a) = \text{CDF}(b) - \text{CDF}(a) = P(a < x < b).$$

- **PDF as a Derivative:**

- What if you only have the PDF $f(x)$? Here's a twist: think of the PDF as the derivative of the CDF. Yep, that's right—PDF is the rate of change of probability for a tiny change in x .

Summing It All Up: The Integral

To find $P(a < x < b)$, sum up all those tiny changes in probability (that is, integrate the PDF):

$$P(a < x < b) = \int_a^b f(x) dx = F(b) - F(a).$$

Voilà! Integrals save the day by summing infinite tiny contributions!

Expectation as the True Mean: The Center of Mass

Def. (True Mean (Expectation) of X) The expectation $\mathbb{E}[X]$ represents the true mean of a random variable X . It is defined as:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

where $f(x)$ is the probability density function (PDF) of X .

Intuition: Center of Mass

- Think of the expectation as the “center of mass” of the probability distribution.
- Just as the center of mass is the point where the mass of a physical object is balanced, $\mathbb{E}[X]$ is the point where the distribution of probabilities balances out.
- This center balances all the possible values of X , weighted by their relative likelihoods (given by $f(x)$).

Numerical Integration

Numerical Integration: The Basics

- **Purpose:**

- Approximate the integral $\int_a^b f(x) dx$ when an analytical solution is difficult or impossible.

- **Key Concept:**

- Break the interval $[a, b]$ into n small subintervals of equal width $\Delta x = \frac{b-a}{n}$.
- Approximate the area under the curve $f(x)$ on each subinterval.

- **Basic Approach:**

- The integral is approximated by summing up these small areas:

$$\int_a^b f(x) dx \approx \sum_{i=1}^n f(x_i) \cdot \Delta x$$

- x_i represents the chosen point in the i -th subinterval.

Example: Trapezoidal Rule for $\int_0^1 e^{-x^2} dx$

- **Goal:** Approximate $\int_0^1 e^{-x^2} dx$ using the Trapezoidal Rule with $n = 4$ subintervals.
- **Step 1: Subinterval Width:** $\Delta x = \frac{1-0}{4} = 0.25$.
- **Step 2: Function Values:** $f(0) = 1$, $f(0.25) \approx 0.939$, $f(0.5) \approx 0.778$, $f(0.75) \approx 0.569$, $f(1) \approx 0.368$
- **Step 3: Apply Trapezoidal Rule:**

$$\int_0^1 e^{-x^2} dx \approx \frac{0.25}{2} [1 + 2(0.939 + 0.778 + 0.569) + 0.368]$$

$$\int_0^1 e^{-x^2} dx \approx 0.743$$

Monte Carlo Integration: Step by Step

- **Step 1: Define the Integral.** Let $\int_a^b f(x) dx$ be the integral you want to approximate.
- **Step 2: Generate Random Samples.** Generate N random points $x_1, x_2, \dots, x_N$ uniformly in the interval $[a, b]$.
- **Step 3: Evaluate the Function.** Compute $f(x_i)$ for each random sample x_i .
- **Step 4: Compute the Average.** Calculate the average value of $f(x)$ over all samples:

$$\text{Average} = \frac{1}{N} \sum_{i=1}^N f(x_i)$$

- **Step 5: Estimate the Integral.** Multiply the average by the interval length $(b - a)$ to estimate the integral:

$$\int_a^b f(x) dx \approx (b - a) \cdot \text{Average}$$

Why Calculate the Average?

Intuition Behind the Average:

- To estimate the area under $f(x)$ over $[a, b]$, think of the average height of $f(x)$ across this interval.
- The average of function values at sampled points gives a representative height:

$$\text{Average} = \frac{1}{N} \sum_{i=1}^N f(x_i)$$

- This average height, when multiplied by the interval length $(b - a)$, provides an approximation of the area:

$$\text{Approximate Area} = (b - a) \times \text{Average}$$

- If the samples are well-distributed, the average closely matches the true mean height of the function, leading to a good approximation of the integral.